

Threshold Lab: Workflow & System Architecture

Calibrating the Sorites Paradox in Credit Card Fraud Detection | COMP702 M.Sc. Project Proposal

1. Executive Summary & Core Objective

Threshold Lab is an interactive web-based machine learning framework designed to demonstrate that the binary decision boundary between 'Genuine' and 'Fraudulent' credit card transactions is a **constructed boundary** derived from cost assumptions, rather than a fixed fact. By applying Eubulides' classical *Sorites Paradox* and Wright's (1975) *tolerance principle* alongside Elkan's (2001) cost-sensitive expected-loss formula, the application allows stakeholders to adjust financial cost ratios live and observe the movement of the optimal decision boundary τ^* , confusion matrices, and the resulting *Contested Zone*.

2. User Interactive Workflow (Step-by-Step)

The end-user interacts with the dashboard through a structured 5-stage workflow:

- **Step 1: Configuration & Dataset Selection**

User selects between the **Synthetic ULB Benchmark Dataset** (50,000 samples, 0.172% fraud prevalence) or uploads a custom **Kaggle creditcard.csv** dataset via the sidebar uploader.

- **Step 2: Model & Calibration Choice**

User selects the classifier architecture (*Logistic Regression* or *Random Forest Ensemble*) and active probability calibration method (*Raw Scores*, *Platt Scaling*, or *Isotonic Regression*).

- **Step 3: Philosophical Sorites Exploration**

User inspects a live worked example of twin transactions straddling τ^* . Although their risk scores differ by less than 0.001, one transaction is silently approved while the other triggers a card block.

- **Step 4: Live Cost-Sensitive Tuning**

User adjusts the sliders for False Positive Cost **C(FP)** (£1 to £200) and False Negative Cost **C(FN)** (£100 to £5000) to observe real-time updating of τ^* , expected monetary loss, PR curves, and confusion heatmaps.

- **Step 5: Contested Zone & CSV Export**

User sets a plausible range of cost assumptions [$C(FP)_{min}$, $C(FP)_{max}$] to quantify all transactions whose label flips, downloading the resulting contested transaction table as a CSV file.

3. Technical System Architecture & Execution Flow

Under the hood, data and computation flow across four modular software layers:

Layer	Module / File	Primary Responsibilities
1. Data Layer	src/data_loader.py	Generates 31 benchmark features (Time, Amount, V1-V28 PCA), performs stratified 80/20 train/test split, and applies standard scaling.
2. ML & Calibration	src/ml_engine.py	Trains baseline models, applies 3-fold Platt scaling (Sigmoidal) & Isotonic Regression, and calculates PR-AUC & Brier scores.
3. Sorites Engine	src/sorites_engine.py	Computes Elkan's τ^* , evaluates confusion matrices & monetary loss, filters the Contested Zone, and finds twin sorites pairs.
4. UI & Dashboard	app.py	Renders dark-themed Streamlit interface, LaTeX equations, Plotly PR curves, reliability diagrams, and CSV export handlers.

4. Key Mathematical Formulations

1. Elkan's Optimal Cost-Sensitive Threshold Formula (Elkan, 2001):

$$\tau^* = C(FP) / [C(FP) + C(FN)]$$
2. Expected Total Financial Loss Function:

$$Loss(\tau) = FP(\tau) \cdot C(FP) + FN(\tau) \cdot C(FN)$$
3. Platt Sigmoidal Probability Scaling (Platt, 1999):

$$P(Y = 1 \mid s) = 1 / [1 + \exp(A \cdot s + B)]$$
4. Contested Zone Transaction Set:

$$C_zone = \{ x \mid \tau^*(R_min) \leq p(x) < \tau^*(R_max) \}$$

5. Author & Academic Context

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